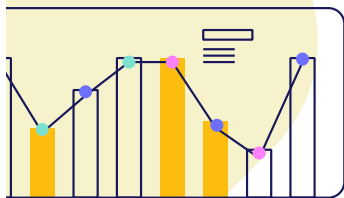
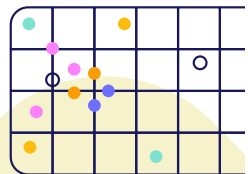


REM2 Research Methods

Lecture 4: Hedonic Price Modelling I



By Helen Bao
Department of Land Economy
University of Cambridge





What will we learn in this lecture?

01.

A crash course on linear regression analysis

Hedonic price modelling (HPM) is an application of multiple linear regression

02.

Applications in real estate research

HPM enjoys wide applications in RE research and beyond

03.

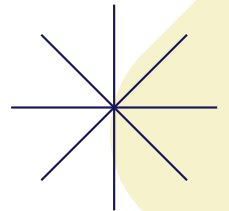
Intercept and slope dummy variables in HPM

A versatile tool in HPM

04.

Exercises

Data transformation, model estimation, diagnostic tests, ...



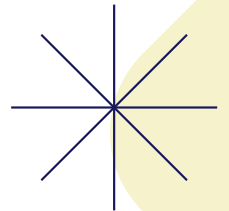


Hedonic price modelling – definition

- A technique to estimate economic values for factors that directly affect market prices
- Usually applied to durable goods (e.g., car and house prices)
- An application of the multiple linear regression method
- Standard functional form:

$$Price = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \cdots + \beta_k X_k + \varepsilon$$

- Interpretation:
 - Price is not directly observable
 - Price is determined by attributes (i.e., Xs in the equation above)
 - β indicates the economic values of the corresponding attribute
 - Price is found by adding up all attributes' economic values



Hedonic price modelling – How to estimate β s?

- Multiple linear regression: the technique is called **Ordinary Linear Squares (OLS)**
- Population model (**parameters** unknown):

$$Price = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_k X_k + \varepsilon$$

- Sample model (**coefficients** are estimated from samples):

$$\widehat{Price} = b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3 + \dots + b_k X_k$$

- The difference between observed values ($Price$) and predicted values ($\widehat{Price}$) is the estimated **residuals** (e). Our objective is to minimize these predictive errors across all observations

$$e = Price - \widehat{Price} = Price - b_0 - b_1 X_1 - \dots$$

$$\text{Min} \sum e^2 = \text{Min} \sum (Price - b_0 - b_1 X_1 - \dots)^2$$

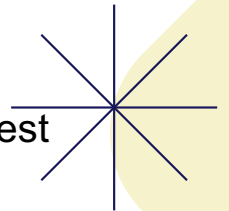
- Leave this to statistical software packages





Hedonic price modelling – How to estimate β s?

- The main technical challenges facing HMP (typical when dealing complex goods):
- Omitted variable biases – when important variables are missing from the regression model
 - Consequences: biased/incorrect coefficient estimates
 - Remedies: robustness checks, no direct solutions (no data!)
- Misspecification errors – when variables are included in wrong forms
 - Consequences: biased/incorrect coefficient estimates
 - Remedies: variable transformations, more flexible estimation methods, ...
- Multicollinearity – when included variables are highly correlated
 - Consequences: inflated variances, biased/incorrect coefficient estimates and test results
 - Remedies: removing correlated variables, orthogonalization, ...



Example 1: The value of transportation accessibility

- N = 12,915
- $R^2 = 0.864$
- Study region: Montgomery County, Maryland, USA
- Hedonic price model:

$$\ln P_{it} = \alpha_t + \theta_t d_t + \mathbf{X}\boldsymbol{\beta} + u_{it}$$

d_t : Distance to transit station at time t (mile)

$\mathbf{X}$: Housing attributes

2011: Announcement year

- Repeat sales model:

$$\ln P_{it} - \ln P_{is} = (\alpha_t + \theta_t d_{it} + \mathbf{X}\boldsymbol{\beta} + u_{it}) - (\alpha_s + \theta_s d_{is} + \mathbf{X}\boldsymbol{\beta} + u_{is})$$

$$(\ln P_{it} - \ln P_{is}) = (\alpha_t - \alpha_s) + (\theta_t - \theta_s) d_i + (u_{it} - u_{is})$$

Variable	Hedonic (anticipated new stations) (1)		Hedonic (established stations) (2)		Repeat-sales (anticipated new stations) (3)		Repeat- sales (established stations) (4)	
	Coef.	SE	Coef.	SE	Coef.	SE	Coef.	SE
Constant	7.828***	0.192	7.021***	0.410				
Logarithm of lot size	0.104***	0.004	0.085***	0.011				
Logarithm of building area	0.364***	0.007	0.339***	0.015				
Age	-0.0005***	0.0001	-0.000005	0.0002				
More than one story	0.013**	0.004	-0.027*	0.011				
Brick	0.005	0.003	0.017*	0.008				
Basement	0.096***	0.005	0.068***	0.011				
Grade of structure	0.146***	0.003	0.125***	0.006				
Logarithm of school quality	0.046	0.038	0.211**	0.074				
Logarithm of median household income	0.093***	0.006	0.159 ***	0.019				
Proportion vacant	-0.006	0.077	0.548	0.302				
Station parking	0.170***	0.009						
Distance to transit station (2002–2011)	0.042***	0.003	-0.122***	0.004	-0.0589	0.154	-0.0393	0.315
Distance to transit station (2012–2017)	-0.032***	0.003	-0.115***	0.006	-0.0949 [†]	0.0595	-0.0787	0.178
Control time fixed effect	Yes		Yes		Yes		Yes	
Control station area difference	Yes		Yes		Yes		Yes	
Test for equality of coefficient of distance	17.44		0.970		37.27		0.012	
	Reject null hypothesis		Cannot reject null hypothesis		Reject null hypothesis		Cannot reject null hypothesis	
No. of observations	12,915		2,345		2,660		427	
R ²	.864		.809		.714		.606	

Note: The natural logarithm of the sales price serves as the dependent variable. The regression includes sixteen dummy variables indicating the year of sale and a series of dummy variable indicating station difference.

[†]p = .1 (two-tailed). *p = .05 (two-tailed). **p = .01 (two-tailed). ***p = .001 (two-tailed).

Source: Peng, Q., and G. Knaap (2023). "When and Where Do Home Values Increase in Response to Planned Light Rail Construction?" *Journal of Planning Education and Research*.

Example 2: The costs of smoking in the rental housing

	(1)	(2)	(3)
<i>No smoking</i>	-0.046*** (0.001)	-0.030*** (0.001)	-0.026*** (0.001)
Property characteristics			
Size			
<i>Number of bedrooms</i>		0.216*** (0.000)	0.233*** (0.000)
<i>Number of floors</i>		-0.043*** (0.000)	-0.032*** (0.000)
<i>Number of bathrooms</i>		0.041*** (0.000)	0.038*** (0.000)
Status			
<i>Refurbished</i>			0.019*** (0.001)
<i>Furnished</i>			0.014*** (0.000)
Type			
<i>Terraced house</i>			-0.083*** (0.001)
<i>Flat</i>			0.053*** (0.001)
<i>Other</i>			-0.010*** (0.001)
Observations	2,726,268	2,726,268	2,726,268
R ²	0.542	0.731	0.737

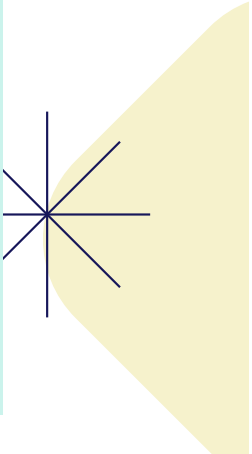
- N = 2,726,268
- R² = 0.737
- Study region/country: UK
- Hedonic price model:
 - $\ln(R_i) = \alpha + \beta \text{No_Smoking}_i + X\gamma + D\delta + \varepsilon_i$
 - No_Smoking = 1 if smoking is not allowed in the rental property
 - X is a matrix of housing attributes; D is a matrix of country and time dummies
 - β captures the rent premium

Source: Gedikli, C., R. Hill, O. Talavera, and O. Yilmaz (2023). "The Hidden Cost of Smoking: Rent Premia in the Housing Market." *Real Estate Economics* 51, no. 3: 611-29.



Hedonic price modelling – Examples



- Why Price (Y) was logged in both models?
 - How to interpret the coefficient estimates of the distance variables in Example 1?
 - How to interpret the coefficient estimates of the smoking indicator (dummy variable) in Example 2?
 - What does the p-value of a t test in these regression outputs tell us?
 - Why a higher R^2 (usually ≥ 0.7) is expected in HPM?
 - Why multiple models were estimated?
 - Example 1: with an alternative estimation method (repeat sales)
 - Example 2: with different model specifications (with different sets of control variables)
- 
- 

Why Price (Y) was logged in both models?

Logging Data – interpretation of the beta coefficients

- When Y is logged, the Y changes by $100 \times (\text{beta coefficient})$ percent for a one unit increase in X

$$\ln(Y) = 8.521 + 0.0055X \quad \text{beta} = 0.0055$$

When X increases by 1 unit, Y increases by 0.55%.

- When X is logged, a one percent change in X changes the dependent variable by $(\text{beta}/100)$ units

$$Y = 12.143 + 1.15 \ln(X) \quad \text{beta} = 1.15$$

When X increase by 1%, Y increases by 0.0115 units or 1.15%.

- When both X and Y are logged, beta is the percent change in Y, while X increases by one percent. This relationship is commonly referred in econometrics as elasticity.

$$\ln(Y) = 2.143 + 0.05 \ln(X) \quad \text{beta} = 0.05$$

When X increase by 1%, Y increases by 0.05%.

How to interpret the coefficient estimates?

Example 1:

$$\ln(\widehat{Price}) = 7.828 + 0.042 \times Distance_to_transit_station_(2002 - 2011) - 0.032 \times Distance_to_transit_station_(2012 - 2017) + \dots$$

- Before 2011 (the announcement year), when distance to transit station increase by one mile (further away), house prices increased by 4.2%, indicating undesirable locations
- After 2011 (the announcement year), when distance to transit station increase by one mile (further away), house prices decreased by 3.2%, indicating desirable locations

What does the p-value of a t test tell us?

Example 1:

$$H_0: \beta_{Distance_to_transit_station_(2002-2011)} = \beta_{Distance_to_transit_station_(2012-2017)}$$

- Reject H_0 if p-value is less than α ($= 0.10, 0.05, 0.01, \text{ or } 0.001$)
- Capitalisation of new infrastructure happened well before the construction started

How to interpret the coefficient estimates?

Example 2:

$$\ln(\widehat{Price}) = constant - 0.026 \times No_Smoking + \dots$$

- When No_Smoking changes its value from 0 to 1, rent decreases by 2.6%.
- The rent premium is estimated to be 2.6%. As the mean weekly rent is around £240 in the sample, this premium corresponds to an average of £6.24 pounds per week.

What does the p-value of a t test tell us?

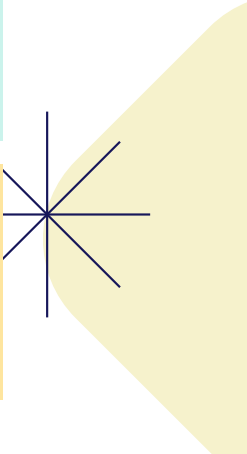
Example 2:

$$H_0: \beta_{No_Smoking} = 0$$

- Reject H_0 if p-value is less than α ($= 0.10, 0.05, 0.01, \text{and } 0.001$)
- Landlords charge smokers differently (for more)



Hedonic price modelling – Examples

- 
- Why a higher R^2 (usually ≥ 0.7) is expected in HPM?
 - Omitted variable bias
 - Predictive accuracy
 - Why multiple models were estimated?
 - Omitted variable bias
 - Misspecification errors
 - Multicollinearity
- 
- 
- These are standard statistics to be reported in a typical HPM analysis
 - Also supplemented with RESET test (misspecification), WHITE test (heteroskedasticity) and VIF checks (multicollinearity).
- 
- 

Dummy variables in HPM

- Linear regression models require numerical input, so categorical variables (such as gender, property type, region) need to be converted into a numerical form.
- Dummy variables (also known as indicator variables or binary variables) take on values of 0 or 1 to indicate the absence (0) or presence (1) of a particular categorical effect.
- Dummy variables are used in regression analysis to convert a categorical variable into a group of dummy variables, which are collectively exhaustive (nothing left out) and mutually exclusive (no overlapping).
- For a variable with p categories, only $p-1$ dummy variables are needed in regression. The omitted category is the reference category.
- Each coefficient (β) of a dummy variable represents the difference in the dependent variable (e.g., rent, price) between the reference category and the respective category of the dummy variable.

Dummy variables in HPM

The original variable is

Property type

- = 1 if flat/apartment
- = 2 if terraced house
- = 3 if semi-detached house
- = 4 if bungalow
- = 5 if detached house

The regression model for the rental market is

$$\widehat{Price} = 601.96 + 8.32 \times Floor_Area + 73.66 \times Property_Type$$

The rental difference between a terraced house (Property type = 2) and a flat (Property type = 1) is £73.66
 The rental difference between a semi-detached house (Property type = 3) and a terraced house (Property type = 2) is £73.66

...

The rental difference between a bungalow (Property type = 4) and a terraced house (Property type = 2) is £73.66 * 2 = £147.32

Regression Summary

	(1) Rental Model	(2) Sales Model
Floor area	8.32*** (49.69)	208.98*** (13.14)
Property type	73.66*** (9.55)	115670.56*** (43.19)
Constant	601.96*** (33.04)	171538.45*** (21.96)
N	4262	6940
R2	0.48	0.25
F	1935.84	1139.76
p	0.00	0.00

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Dummy variables in HPM

Property type should be transformed into a group of dummy variables:

- Flat = 1 if Property type = “flat/apartment” and 0 otherwise. **Reference category**
- Terraced House = 1 if Property type = “terraced” and 0 otherwise.
- Semi-detached House = 1 if Property type = “semi-detached” and 0 otherwise.
- Bungalow = 1 if Property type = “bungalow” and 0 otherwise.
- Detached House = 1 if Property type = “detached” and 0 otherwise.

The regression model for the rental market is

$$\widehat{Price} = 670.99 + 8.20 \times Floor_area + 120.08 \times Terraced_House + 159.51 \times Semi_Detached_House - 39.00 \times Bungalow + 329.55 \times Detached_House$$

The rental difference between a bungalow (Bungalow = 1) and a flat (omitted) is £39.00

The rental difference between a flat (omitted) and a detached house (Detached House = 1) is £329.55

...

The rental difference between a bungalow (Bungalow = 1) and a terraced house (Terraced House = 1) is £120.08+£39.00= £159.08

Regression Summary

	(1) Rental Model	(3) Rental Model
Floor area	8.32*** (49.69)	8.20*** (48.50)
Property type	73.66*** (9.55)	
Terraced House		120.08*** (5.82)
Semi-Detached House		159.51*** (6.81)
Bungalow		-39.00 (-0.54)
Detached House		329.55*** (9.66)
Constant	601.96*** (33.04)	670.99*** (39.58)
N	4262	4262
R2	0.48	0.48
F	1935.84	781.69
p	0.00	0.00

t statistics in parentheses

* p < 0.10, ** p < 0.05, *** p < 0.01

Dummy variables in HPM

Regression Summary

	(1) Rental Model	(3) Rental Model
Floor area	8.32*** (49.69)	8.20*** (48.50)
Property type	73.66*** (9.55)	
Terraced House		120.08*** (5.82)
Semi-Detached House		159.51*** (6.81)
Bungalow		-39.00 (-0.54)
Detached House		329.55*** (9.66)
Constant	601.96*** (33.04)	670.99*** (39.58)
N	4262	4262
R2	0.48	0.48
F	1935.84	781.69
p	0.00	0.00

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Property type:

- = 1 if flat/apartment
- = 2 if terraced house
- = 3 if semi-detached house
- = 4 if bungalow
- = 5 if detached house

Difference vs Flat	Property type	Dummy variables
Terraced House	73.66	120.08
Semi-Detached House	147.32	159.51
Bungalow	220.98	-39.00
Detached House	294.64	329.55

Difference between	Property type	Dummy variables
Flat & Terraced House	73.66	120.08
Terraced House & Semi-Detached House	73.66	39.43
Semi-Detached House & Bungalow	73.66	-198.51
Bungalow & Detached House	73.66	368.55

Dummy variables:

- Bungalow = 1 if Property type = "bungalow" and 0 otherwise.
- Flat = 1 if Property type = "flat/apartment" and 0 otherwise.
- Detached House = 1 if Property type = "detached" and 0 otherwise.
- Semi-detached House = 1 if Property type = "semi-detached" and 0 otherwise.
- Terraced House = 1 if Property type = "terraced" and 0 otherwise.

Which category to omit? It does not matter!

Regression Summary

	(1) Flat	(2) Terraced House	(3) Semi-Detac hed House	(4) Bungalow	(5) Detached House
Floor area	8.20***	8.20***	8.20***	8.20***	8.20***
Terraced House	120.08***		-39.43*	159.07**	-209.48***
Semi-Detached House	159.51***	39.43*		198.50***	-170.04***
Bungalow	-39.00	-159.07**	-198.50***		-368.55***
Detached House	329.55***	209.48***	170.04***	368.55***	
Flat/Apartment		-120.08***	-159.51***	39.00	-329.55***
Constant	670.99***	791.07***	830.50***	632.00***	1000.54***
N	4262	4262	4262	4262	4262
R2	0.48	0.48	0.48	0.48	0.48
F	781.69	781.69	781.69	781.69	781.69
p	0.00	0.00	0.00	0.00	0.00

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

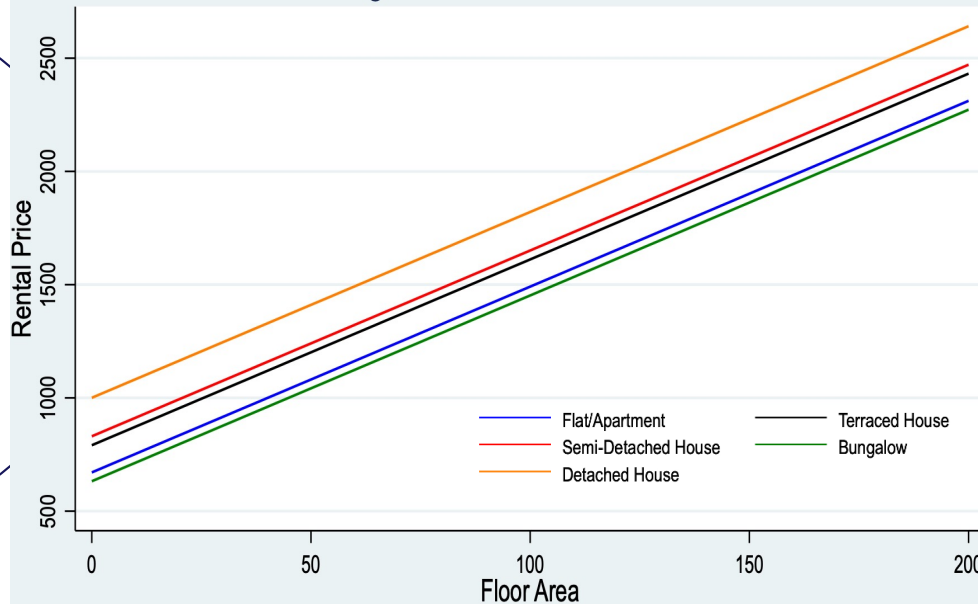
Intercept Dummy Variables

A group of dummy variables defines a group of parallel regression lines. The differences among these regression lines are their intercept terms.

The regression model for the rental market is

$$\begin{aligned}\widehat{Price} &= 670.99 + 8.20 \times Floor_area + 120.08 \times Terraced_House \\ &+ 159.51 \times Semi_Detached_House - 39.00 \times Bungalow \\ &+ 329.55 \times Detached_House\end{aligned}$$

Parallel Regression Lines for the Rental Model



This is a combination of five models:

- When Flat = 1

$$\widehat{Price} = 670.99 + 8.20 \times Floor_area$$

- When Terraced House = 1

$$\widehat{Price} = 670.99 + 8.20 \times Floor_area + 120.08 = 791.07 + 8.20 \times Floor_area$$

- When Semi-detached House = 1

$$\widehat{Price} = 670.99 + 8.20 \times Floor_area + 159.51 = 830.50 + 8.20 \times Floor_area$$

- When Bungalow = 1

$$\widehat{Price} = 670.99 + 8.20 \times Floor_area - 39.00 = 631.99 + 8.20 \times Floor_area$$

- When Detached House = 1

$$\widehat{Price} = 670.99 + 8.20 \times Floor_area + 329.55 = 1000.54 + 8.20 \times Floor_area$$

Slope Dummy Variables

- The slope of a regression line is changed by introducing interaction terms with dummy variables.
- For example, the effect of an independent variable (e.g., floor area) on the dependent variable (e.g., rent) varies across different groups represented by a dummy variable (e.g., flat)

$$Price = \beta_0 + \beta_1 \times \text{Floor_area} + \beta_2 \times \text{Terraced_House} + \beta_3 \times \text{Semi_Detached_House} + \beta_4 \times \text{Bungalow} + \beta_5 \times \text{Detached_House} + \beta_6 \times \text{Detached_House} \times \text{Floor_area} + \varepsilon$$

- For detached houses (i.e., $\text{Detached_House} = 1$), the regression model is

$$Price = (\beta_0 + \beta_5) + (\beta_1 + \beta_6) \times \text{Floor_area} + \beta_2 \times \text{Terraced_House} + \beta_3 \times \text{Semi_Detached_House} + \beta_4 \times \text{Bungalow} + \varepsilon$$

- For other types of properties (i.e., $\text{Detached_House} = 0$), the regression model is

$$Price = \beta_0 + \beta_1 \times \text{Floor_area} + \beta_2 \times \text{Terraced_House} + \beta_3 \times \text{Semi_Detached_House} + \beta_4 \times \text{Bungalow} + \varepsilon$$

Slope Dummy Variables

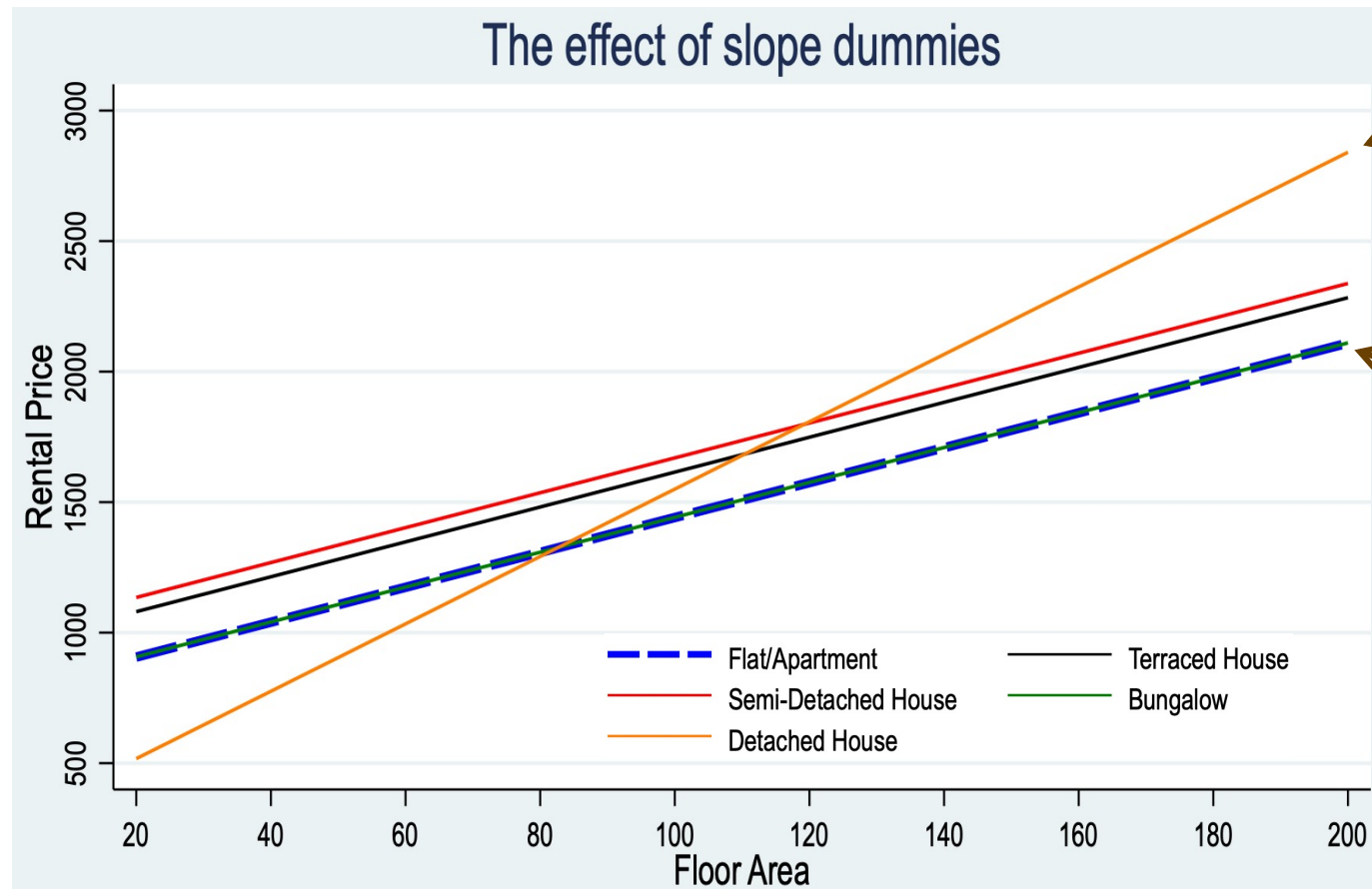
A 'size premium' for detached houses

Regression Summary

	(1) Intercept dummies only	(2) Intercept & slope dummies	(3) Intercept dummies only	(4) Intercept & slope dummies
Floor area	8.20***	6.68***	203.06***	112.08***
Terraced House	120.08***	173.92***	152571.04***	156654.53***
Semi-Detached House	159.51***	228.37***	205006.71***	210236.01***
Bungalow	-39.00	0.77	163145.92***	169690.54***
Detached House	329.55***	-513.57***	512045.11***	419792.65***
Detached House # Floor area		6.22***		549.67***
Constant	670.99***	772.78***	282479.24***	288780.18***
N	4262	4262	6940	6940
R ²	0.48	0.51	0.27	0.28
F	781.69	736.13	504.83	460.00
p	0.00	0.00	0.00	0.00

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Slope Dummy Variables

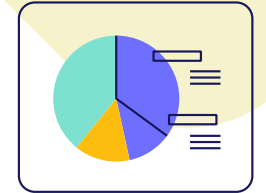


A 'size premium' for larger detached houses, but a discount for smaller ones.

No differences between flats and bungalows

Exercises – rental vs sales markets

1. Open “Lectures 4-5 Exercises Data.xlsx” in a software of your choice. Examine the data.
2. Generate descriptive statistics of all variables. Note the distribution of ‘Floorarea’. What is the maximum value of this variable? Remove observations with a floor area over 200 squared metres.
3. Make natural log transformation of the following variables: Price, Floorarea, Busstation, Railstation and Airport. Save the file to your work directory.
4. Convert Propertytype and Date into dummy variables.
5. Estimate a hedonic price model for the rental and the sales market, respectively. Include intercept dummies for property type and transaction year. Use both the linear and the log-linear functional forms.
6. Estimate a hedonic price model for the rental market only. Include a slope dummy to check if the effect of Floorarea is different between Detached houses and other types of properties.



Further Questions?